ISEL | DEI | AMD / MLDM

Machine Learning and Data Mining

1. Context and objective (*just read; do nothing yet!*)

This MoP continues directly from "MoP_02 (From Data Source to Dataset — 1 of 2)".

The previous MoP_02 integrated the collected penguin observations into a relational database (PostgreSQL) and completed the missing island information. Its main result was therefore not a new dataset file, but the **persistent state** of the integrated and completed data source.

The current MoP_03 starts from that **persistent state** and progresses towards a **materialized analytical dataset** suitable for later modelling.

The activity of the current MoP has **two main stages**:

- analyse the integrated source and choose the attributes to include in the analytical dataset
- construct, validate and materialize that analytical dataset

The **analysis stage** includes target distribution, contextual and morphological evidence, multidimensional aggregation, attribute provenance and prediction-time availability.

The **construction stage** then applies an explicit observation-eligibility rule, validates the intended observation-unit and data-grain, examines the effect of eligibility on the target distribution, and exports the resulting analytical dataset.

The final analytical dataset will be:

`03_output/penguins_analytical_dataset.csv`

A second Orange Data Mining (DM) checkpoint provides a visual comparison between the original collected source and the materialized analytical dataset, helping to verify the effects of attribute-selection and observation-eligibility decisions on the resulting dataset.

2. Prerequisites for the current MoP (and continuity from previous MoP)

MoP_03 must not be started independently. Complete MoP_02 first.

If the final PostgreSQL state from MoP_02 is no longer available, return to MoP_02 and complete its integration and completion notebooks again before continuing.

The following artefacts from MoP_02 **are required** in the current MoP workspace, so:

- copy `_mLDm_00_penguins_data_collected.csv` into `01_input`
- copy `penguins_visual_check.ows` into `02_code`

These files are required later for the second Orange DM visual checkpoint.

Do not interpret this instruction as a requirement to copy the whole MoP_02 workspace. Preserve the PostgreSQL state and **copy only** the artefacts referred above.

Note 1: the PostgreSQL state is **not a file to copy**; it must remain available in the database container used in MoP_02.

Note 2: do **not** rerun the MoP_02 bootstrap script as part of the current MoP; the bootstrap script recreates the original database state and would therefore discard the integration and completion work required by MoP_03.

3. Working principles

While completing the activity:

- use the integrated PostgreSQL source produced in MoP_02
- inspect evidence before making analytical decisions
- distinguish association with the target from legitimate explanatory information
- consider attribute meaning, provenance and prediction-time availability
- use expected-result checkpoints to detect mistakes, not as substitutes for understanding
- preserve the intended observation-unit and data-grain in the analytical dataset
- keep all original `_mLDm_` files unchanged
- complete all `TODO` sections, evidence answers and conclusions in the notebooks
- place each required artefact in the indicated folder

Before considering the activity complete, verify that:

- both notebooks execute in the required order
- the analytical dataset has been materialized successfully
- the Orange workflow has been extended and saved
- all required student-created or student-modified artefacts are ready for submission

4. Course files and working copies

Files whose names begin with `_mLDm_` are provided by the course and must remain unchanged.

Only files that you are going to modify must be duplicated. For those files:

- duplicate the original `_mLDm_` file
- remove the `_mLDm_` prefix from the duplicate filename
- work only on the duplicate
- keep the original course file unchanged

4.1. Prepare the working copies

In this MoP, the following files in `02_code` must be duplicated and the `_mLDm_` prefix removed:

- `_mLDm_03_penguins_analysis.ipynb`

to `03_penguins_analysis.ipynb`

- `_mLDm_04_penguins_dataset.ipynb`

to `04_penguins_dataset.ipynb`

Keep the original `_mLDm_` files unchanged and work only on the corresponding copies.

5. Files and execution order

5.1. Input files

The `01_input` folder must contain the files required by the current activity, including:

- `_mLDm_00_penguins_data_collected.csv`
- `_mLDm_xx_submission_requirements.txt`

Recall that the data-collected CSV must be copied from MoP_02 and is used again in the current MoP only for the Orange DM comparison checkpoint.

The submission requirements file is provided with this MoP and must not be modified. It is used by the submission tool to validate the required artefacts.

5.2. Code files

The `02_code` folder contains:

- `_mLDm_03_penguins_analysis.ipynb`
- `_mLDm_04_penguins_dataset.ipynb`
- `_mLDm_xx_build_submission_zip.py`

and must also contain the Orange workflow carried from MoP_02:

- `penguins_visual_check.ows`

After creating the notebook working copies, execute them in this order:

1. `03_penguins_analysis.ipynb`
2. `04_penguins_dataset.ipynb`

The notebooks connect directly to the integrated PostgreSQL data source produced in MoP_02.

Do not recreate the initial PostgreSQL database before running them.

The notebooks contain expected-result checkpoints; use them to validate intermediate steps.

6. Part C — Analytical exploration and attribute selection

Open:

`03_penguins_analysis.ipynb`

This notebook builds a flat analytical view over the integrated database and uses it to investigate the target and the available attributes before selecting candidate explanatory attributes.

Complete the `TODO` sections and evidence questions so that you can:

- inspect the target distribution
- construct and interpret a multidimensional `CUBE`
- distinguish source `NULL` values from dimensions aggregated by the cube
- identify theoretically possible but unobserved species × island combinations
- use `PIVOT` to inspect the observed species × island structure
- compare average morphological measurements across species
- compare species composition across collection years
- use window functions to compute within-year percentages
- inspect `sample_number` as evidence about the data-collection process
- inspect the association between batch assignment and species
- inspect laboratory assignment and compare it with the batch pattern
- reason about attribute meaning, provenance and prediction-time availability

The purpose is not simply to identify attributes associated with species, but **to decide**, for each available attribute, whether it should be treated as:

- the target
- a retained candidate explanatory attribute
- an excluded attribute

Give a short justification for every retained or excluded attribute.

Pay particular attention to attributes that may encode source, collection or processing context. Strong association with the target does not by itself make an attribute appropriate for use in the analytical dataset.

The final cell of this notebook records the candidate explanatory attribute set that will be used by notebook `04`.

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7. Part D — Analytical dataset construction

Open:

`04_penguins_dataset.ipynb`

only after notebook `03_penguins_analysis.ipynb` has been completed.

This notebook turns the analytical decisions made in notebook `03` into a reproducible analytical dataset.

The notebook also consolidates two related concepts:

- **observation-unit** - what one record represents conceptually
- **data-grain** - the level of detail at which that observation-unit is represented

In this activity:

- the observation-unit is one observation of one penguin
- the intended data-grain is one specific penguin observed at one specific `date_time`
- therefore, each observation-unit is uniquely identified by `(id_penguin, date_time)`

Note: some attributes may be retained temporarily for validation purposes even if they are not part of the final analytical dataset.

Complete the `TODO` sections and evidence questions so that you can:

- construct the selected analytical source using the decisions from notebook `03`
- quantify missing values in the selected explanatory attributes and target
- define an observation-eligibility rule
- apply a complete-case eligibility rule for this analytical path
- construct the analytical dataset
- validate the number of eligible observations
- validate the intended observation-unit and data-grain
- compare the target distribution before and after applying observation eligibility
- interpret the effect of excluded observations on dataset size and class proportions
- materialize the final analytical dataset as a CSV file

The resulting file is:

`03_output/penguins_analytical_dataset.csv`

Before continuing, verify that this file has been created successfully.

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8. Second visual checkpoint with Orange Data Mining (DM)

The MoP_02 introduced Orange DM using the original collected CSV and created:

`02_code/penguins_visual_check.ows`

In this MoP, do **not** recreate that workflow from scratch.

Reopen and extend the workflow carried from MoP_02 so that it contains two branches:

- the original collected source:

`01_input/_mLDm_00_penguins_data_collected.csv`

- the analytical dataset produced in the current MoP:

`03_output/penguins_analytical_dataset.csv`

Follow the instructions at the end of `04_penguins_dataset.ipynb`.

For the analytical dataset verify that:

- `species` is the **Target**
- the remaining analytical attributes are **Features**
- attributes excluded during the previous analysis are no longer present

Use `DataTable`, `Distributions` and `ScatterPlot` to compare the collected source with the analytical dataset.

Consider in particular:

- which attributes are present
- which roles are assigned to them
- the number and nature of the observations
- the visual distributions and relationships between selected attributes

The purpose of this checkpoint is to compare the **collected source** with the **materialized analytical dataset** and visually inspect the effects of the attribute-selection and observation-eligibility decisions made throughout this MoP.

Do **not** train or compare predictive models yet.

Save the extended workflow again as:

`02_code/penguins_visual_check.ows`

No additional Orange DM screenshot or report artefact is required in this MoP.

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9. Required notebook completion

There is no separate final-evidence cell in this MoP.

Instead, before submission, verify that:

- every required `TODO` section has been completed
- every evidence question has been answered
- every requested conclusion has been written
- the final candidate attribute selection in notebook `03` is complete
- the CQ1, CQ2 and CQ3 conclusions in notebook `04` are complete
- the analytical dataset has been created successfully
- the Orange DM workflow has been extended and saved

Answers should be concise and supported by evidence obtained during notebook execution.

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10. Submission checklist

Before submitting, verify that you have:

- `02_code/03_penguins_analysis.ipynb`
- `02_code/04_penguins_dataset.ipynb`
- `02_code/penguins_visual_check.ows`
- `03_output/penguins_analytical_dataset.csv`

Use `_mLDm_xx_build_submission_zip.py`, as in MoP_02, to validate the required artefacts and create the ZIP file for submission.

The tool uses:

`01_input/_mLDm_xx_submission_requirements.txt`

to determine which artefacts are required.

It also creates:

`04_report/submission_check.txt`

If a required artefact is missing, validation fails and the submission ZIP is not created.

Before uploading the ZIP to Moodle:

- check that `submission_check.txt` reports successful validation
- inspect the ZIP contents
- verify once more that the notebooks contain all required answers and conclusions